NoisePrivacyMap<L1Distance<RBig>, MultiDP> for ZExpFamily<1>

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

This proves the privacy-map implementation for the discrete Laplace mechanism when the output measure is `MultiDP`.

1 Hoare Triple

Precondition

`MI` is `L1Distance<RBig>` and `MO` is `MultiDP`. The distribution scale is a non-negative rational.

Pseudocode

```python
class ZExpFamily1:
    def noise_privacy_map(self, _metric, _measure):
        scale = self.scale
        if scale < RBig.ZERO:  #
            raise "scale must be non-negative"
        if scale == RBig.ZERO:  #
            raise "zero scale is unsupported for MultiDP"
            raise "scale must be non-negative"

        def privacy_map(d_in):
            if d_in < RBig.ZERO:  #
                raise "sensitivity must be non-negative"
            if d_in == RBig.ZERO:  #
                return (
                    PrivacyGuarantee()
                    .with_approxDP([(0.0, 0.0)])
                    .with_zCDP(0.0, 0.0)
                    .with_renyiDP(lambda alpha: 0.0, 0.0)
                )
            epsilon_exact = d_in / scale
            epsilon = f64.inf_cast(epsilon_exact)
            if not epsilon.is_finite():
                raise "privacy parameters are not finite"

            rho = zcdp_discrete_laplace(epsilon_exact, d_in, scale)
            curve = PrivacyGuarantee().with_approxDP([(epsilon, 0.0)]).with_zCDP(rho, 0.0)
            rdp = lambda alpha: (
                rdp_discrete_laplace(alpha, d_in, scale)
                .or_else(lambda: rdp_from_pureDP(alpha, epsilon))
                .or_else(lambda: epsilon)
            )
            return curve.with_renyiDP(rdp, 0.0)

```

```
34          return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. For every input distance accepted by the returned privacy map, the map returns a `PrivacyGuarantee` containing valid privacy representations for adding an independent discrete Laplace sample to each coordinate.

Proof. Lines 4 and 11 reject invalid parameters. With zero sensitivity, line 13 returns the identity guarantee. With zero scale, MultiDP measurement construction rejects the mechanism because positive sensitivity would have no finite guarantee. The defensive check on line 6 reports the same failure if the privacy map is requested directly.

For positive rational sensitivity s and scale b, the discrete Laplace point-mass ratio calculation gives the pure-DP bound

$$\varepsilon = s/b.$$

The upward cast makes the floating-point `epsilon` an upper bound on this value, so the $(\varepsilon, 0)$ point inserted into the guarantee is valid. The zCDP and RDP calculations retain the exact rational sensitivity, scale, and ratio, enclosing each rational with outward-rounded endpoints before using directed interval arithmetic. Therefore the returned ρ and RDP values are no smaller than their exact bounds. Finally, `rdp_discrete_laplace` evaluates the exact Rényi expression with directed intervals. If that evaluation fails, the implementation first uses the exact RDP value implied by pure DP and, if those numerics also fail, uses the elementary pure-DP bound $D_\alpha \leq \varepsilon$. Each result is an upper bound at every valid order. Each representation is therefore simultaneously valid, and querying or composing the resulting `PrivacyGuarantee` preserves conservativeness. ☐